2022080054 杨礼谦 概率论与数理统计

19. You are told that of the four cards face down on the table, two are red and two are black. If you guess all four at random, what is the probability that you get 0, 2, 4 right?

Assume guessing 4 cards at random means we draw 2 and guess they are red and the remaining will be black,

There $\binom{4}{2}$ ways to pick 2 cards out of 4. 6 ways.

$P(0) = \left(\frac{2}{4}\right)\left(\frac{1}{3}\right) = \underline{\frac{1}{6}}$

$P(2) = \frac{4}{4}\left(\frac{2}{3}\right)$ must be diff from first card

$= \frac{2}{3}$

can be any card

$P(4) = \left(\frac{2}{4}\right)\left(\frac{1}{3}\right) = \underline{\frac{1}{6}}$

20. An airport shuttle bus makes 4 scheduled stops for 15 passengers. What is the probability that all of them get off at the same stop? What is the probability that someone (at least one person) gets off at each stop?

$\boxed{A}$ $\boxed{B}$ $\boxed{C}$ $\boxed{D}$

Probability of all passengers getting off at the same bus stop

$= \frac{\binom{4}{1}}{\binom{15+4-1}{4-1}} = \frac{4}{816} = \frac{1}{204}$

Prob that at least one person gets off at each stop

$= ($

21. Ten books are made into 2 piles. In how many ways can this be done if books as well as piles may or may not be distinguishable? Treat all four hypotheses, and require that neither pile be empty.

a) Both distinguishable
ways: 2^n

Both empty piles are prohibited:
$2^{10}-2=1022$

b) Only books are distinguishable
$\frac{2^{10}-2}{2}=511$

(divide by 2 because other doesn't matter)

c) Only piles are distinguishable
$10-1=9$

d) Both are not distinguishable
5

(1,9), (2,8), (3,7), (4,6), (5,5)

25. One hundred trout are caught in a little lake and returned after they are tagged. Later another 100 are caught and found to contain 7 tagged ones. What is the probability of this if the lake contains n trout? [What is your best guess as to the true value of n? The latter is the kind of question asked in *statistics*.]

$n \rightarrow 100$

$$P=\frac{\binom{n-100}{93}\binom{100}{7}}{\binom{n}{100}}$$

For the best guess, there are 7 tagged when we randomly choose 100 trouts from the lake. We can suppose that 100 trout represent 7% of the population.

$\therefore \quad 100\times\frac{100}{7}=1428.57\approx 1429$

*28. Show that

$$\binom{2n}{n} = \sum_{k=0}^{\infty} \binom{n}{k}^2.$$

[Hint: apply (3.3.9).]

$$\binom{2n}{n} = \sum_{k=0}^{n} \binom{n}{k}\binom{2n-n}{n-k}$$

$$= \sum_{k=0}^{n} \binom{n}{k}\binom{n}{n-k}$$

since $\binom{n}{k} = \binom{n}{n-k}$

then $\sum_{k=0}^{n} \binom{n}{k}\binom{n}{n-k} = \sum_{k=0}^{n} \binom{n}{k}\binom{n}{k}$

$$= \sum_{k=0}^{n} \binom{n}{k}^2$$

$\binom{2n}{n} = \sum_{k=0}^{n} \binom{n}{k}^2$ (Shown)

1. If X is a random variable [on a countable sample space], is it true that

$$X + X = 2X, X - X = 0?$$

Explain in detail.

No it is not true. Take X = outcomes of die rolls, $X_1 + X_2$ = the number of the two rolls which can be either odd or even but $2X$ will always be even. Hence $X + X \neq 2X$ if X is a random variable on a countable sample space.

2. Let $\Omega = \{\omega_1, \omega_2, \omega_3\}$, $P(\omega_1) = P(\omega_2) = P(\omega_3) = 1/3$, and define X, Y, and Z as follows:

$$X(\omega_1) = 1, X(\omega_2) = 2, X(\omega_3) = 3;$$

$$Y(\omega_1) = 2, Y(\omega_2) = 3, mY(\omega_3) = 1;$$

$$Z(\omega_1) = 3, Z(\omega_2) = 1, Z(\omega_3) = 2.$$

Show that these three random variables have the same probability distribution. Find the probability distributions of $X + Y$, $Y + Z$, and $Z + X$.

$P(X=1) = P(\omega_1) = \frac{1}{3}$	$P(Y=1) = P(\omega_3) = \frac{1}{3}$	$P(Z=1) = P(\omega_2) = \frac{1}{3}$
$P(X=2) = P(\omega_2) = \frac{1}{3}$	$P(Y=2) = P(\omega_1) = \frac{1}{3}$	$P(Z=2) = P(\omega_3) = \frac{1}{3}$
$P(X=3) = P(\omega_3) = \frac{1}{3}$	$P(Y=3) = P(\omega_2) = \frac{1}{3}$	$P(Z=3) = P(\omega_1) = \frac{1}{3}$

Hence, all three random variables (X, Y, Z) have the same probability distribution.

$P(X+Y=3) = P(\omega_1) = \frac{1}{3}$	$P(Y+Z=3) = P(\omega_3) = \frac{1}{3}$	$P(Z+X=3) = P(\omega_2) = \frac{1}{3}$
$P(X+Y=4) = P(\omega_3) = \frac{1}{3}$	$P(Y+Z=4) = P(\omega_2) = \frac{1}{3}$	$P(Z+X=4) = P(\omega_1) = \frac{1}{3}$
$P(X+Y=5) = P(\omega_2) = \frac{1}{3}$	$P(Y+Z=5) = P(\omega_1) = \frac{1}{3}$	$P(Z+X=5) = P(\omega_3) = \frac{1}{3}$

Thus, the probability distributions of $X+Y$, $Y+Z$, and $Z+X$ are also the same.

3. In No. 2 find the probability distribution of

$$X + Y - Z, \sqrt{(X^2 + Y^2)Z}, \frac{Z}{|X - Y|}.$$

$P(X+Y-Z=0) = P(\omega_1) = \frac{1}{3}$ $\quad$ $P(\sqrt{(X^2+Y^2)Z} = \sqrt{13}) = P(\omega_2) = \frac{1}{3}$

$P(X+Y-Z=2) = P(\omega_3) = \frac{1}{3}$ $\quad$ $P(\sqrt{(X^2+Y^2)Z} = \sqrt{15}) = P(\omega_1) = \frac{1}{3}$

$P(X+Y-Z=4) = P(\omega_2) = \frac{1}{3}$ $\quad$ $P(\sqrt{(X^2+Y^2)Z} = \sqrt{20}) = P(\omega_3) = \frac{1}{3}$

$P(\frac{Z}{|X-Y|} = 3) = P(\omega_1) = \frac{1}{3}$

$P(\frac{Z}{|X-Y|} = 1) = P(\omega_1) + P(\omega_2) = \frac{2}{3}$

4. Take Ω to be a set of five real numbers. Define a probability measure and a random variable X on it that takes the values $1, 2, 3, 4, 5$ with probabilities $1/10, 1/10, 1/5, 1/5, 2/5$ respectively; another random variable Y that takes the value $\sqrt{2}, \sqrt{3}, \pi$ with probabilities $1/5, 3/10, 1/2$. Find the probability distribution of XY. [Hint: the answer depends on your choice and is not unique.]

$\Omega = \{\omega_1, \omega_2, \omega_3, \omega_4, \omega_5\}$

$P(\omega_1) = \frac{1}{10}$ $\quad$ $X(\omega_1) = 1$ $\quad$ $Y(\omega_1) = \sqrt{2}$

$P(\omega_2) = \frac{1}{10}$ $\quad$ $X(\omega_2) = 2$ $\quad$ $Y(\omega_2) = \sqrt{3}$

$P(\omega_3) = \frac{1}{5}$ $\quad$ $X(\omega_3) = 3$ $\quad$ $Y(\omega_3) = Y(\omega_4) = Y(\omega_5) = \pi$

$P(\omega_4) = \frac{1}{5}$ $\quad$ $X(\omega_4) = 4$

$P(\omega_5) = \frac{2}{5}$ $\quad$ $X(\omega_5) = 5$

$P(XY = \sqrt{2}) = P(XY = 2\sqrt{3}) = P(\omega_1) = P(\omega_2) = \frac{1}{10}$

$P(XY = 3\pi) = P(XY = 4\pi) = P(\omega_3) = P(\omega_4) = \frac{1}{5}$

$P(XY = 5\pi) = P(\omega_5) = \frac{2}{5}$

11. Let $\lambda > 0$ and define f as follows:

$$f(u) = \begin{cases} \frac{1}{2}\lambda e^{-\lambda u} & \text{if } u \geq 0; \\ \frac{1}{2}\lambda e^{+\lambda u} & \text{if } u < 0. \end{cases}$$

This f is called *bilateral exponential*. If X has density f, find the density of $|X|$. [Hint: begin with the distribution function.]

let $Y=|X|$ since $Y\geq 0$, we only need to consider $u\geq 0$

distribution function of Y: $F_Y(u)=P(Y\leq u)$

$=P(|X|\leq u)$

$=P(-u\leq X\leq u)$

$=P(-u\leq X\leq 0)+$

$P(0\leq X\leq u)$

$=\int_{-u}^{0}\frac{1}{2}\lambda e^{-\lambda(-u)}du+\int_{0}^{u}\frac{1}{2}\lambda e^{-\lambda u}du$

$=\int_{0}^{v}\frac{1}{2}\lambda e^{-\lambda v}dv+\int_{0}^{u}\frac{1}{2}\lambda e^{-\lambda u}du$

$=\int_{0}^{u}\lambda e^{-\lambda u}du$ let $v=-u$

density of $|X|$: $F_Y'(u)=\lambda e^{-\lambda u}$ for $u\geq 0$

15. Suppose that

$$p_n = cq^{n-1}p, \quad 1 \le n \le m,$$

where c is a constant and m is a positive integer; cf. (4.4.8). Determine c so that $\sum_{n=1}^{m} p_n = 1$. (This scheme corresponds to the waiting time for a success when it is supposed to occur within m trials.)

$\sum_{n=1}^{m}p_n=\sum_{n=1}^{m}cq^{n-1}p=cp\sum_{n=1}^{m}q^{n-1}$

$1=cp\sum_{n=1}^{m}q^{n-1}$ $\frac{1}{cp}=\sum_{n=1}^{m}q^{n-1}$ (geometric progression)

$$\sum_{n=1}^{m} q^{n-1} = \frac{1-q^m}{1-q}$$

$$\frac{1}{cp} = \frac{1-q^m}{1-q}$$

$$1-q = (1-q^m)cp$$

$$1-q = c(p-pq^m)$$

$$c = \frac{1-q}{p(1-q^m)}$$